

1 Unknown-Distribution, Unknown-Cost Pandora Stopping

1.1 Problem definition

For one prompt or programming problem, consider an exchangeable stream of candidate generations (the “boxes”). Opening box j reveals a raw score R_j and its character length C_j . Before opening it, neither realization is known. More strongly, the prompt-specific reward law and the distribution of the next character length are unknown. After n openings the observable history and decision-space incumbent are

$$\mathcal{H}_n = \{(R_j, C_j) : j \leq n\}, \quad V_n = \max_{j \leq n} u_n(R_j), \quad (1)$$

where u_n converts a raw score into the policy’s estimate of utility. Let $\mathcal{U}_x(j)$ denote the actual evaluation utility of candidate j for task x (BT quality in alignment and binary correctness in coding). Given a character-cost divisor D , the evaluation objective is

$$\mathbb{E} \left[\mathcal{U}_x(J_\tau) - \frac{1}{D} \sum_{j=1}^{\tau} C_j \right], \quad (2)$$

where τ is a stopping time and J_τ is the selected opened candidate. Thus $1/D$ is the externally specified price of one character in utility units; it is not the unknown quantity. The unknown cost is the length of the next generation. Larger D makes characters cheaper and normally induces more openings.

For a known, homogeneous value law F and known constant opening cost c , classical Pandora stopping can be written in expected-improvement form: continue from incumbent v exactly when

$$\mathbb{E}_{X \sim F} [(u(X) - v)_+] > c. \quad (3)$$

Equivalently, the reservation value z solves $\mathbb{E}[(u(X) - z)_+] = c$. Our setting replaces both sides of Eq. (3): an optimistic or regularized upper-tail model is fit from training data and the opened prefix, and the next character cost is estimated from the same information. Because all future generations from a given generator are treated as exchangeable, there is no ordering problem among heterogeneous boxes; the decision is only whether to open one more.

1.2 Utility–cost calibration

The raw reward and the character count cannot be compared directly. The two implementations put them into common units in two steps:

1. *Value calibration* maps the raw reward into an interpretable utility. Alignment uses a Bradley–Terry (BT) probability of beating an online-estimated high-quality benchmark. Coding uses an isotonic estimate of the probability that a program is correct.
2. *Cost calibration* divides the estimated next character count by D . The stopping comparison is therefore “expected utility improvement” versus “expected character cost in utility units.” The realized total cost reported by every experiment is the exact sum of the lengths actually opened, not this estimate.

The phrase *held-out calibration* requires care. In coding, the isotonic map is fit on all examples from the outer-training problems and then frozen; it is held out from the final outer test, but it is not refit inside each fold of the four-fold policy-selection calculation. In alignment, no labeled held-out utility map is fit: the BT benchmark is recomputed from the currently opened prefix. A full-pool benchmark is used only to score a stopped policy after the fact and never enters its decisions. Consequently, the claim that the two value transforms are “held-out calibration and BT” is correct at a high level only with these qualifications.

1.3 A shared high-level algorithm

Let $\hat{F}_n^+$ be the fitted optimistic law of a future value, let $\hat{C}_n$ be the fitted mean length of the next generation, and let A_n and B_n denote method-specific multiplicative and additive regularizers. Both implementations instantiate Algorithm 1.

The stop inequality is non-anticipating: although the experiment code vectorizes calculations over all prefixes, the entry used at prefix n depends only on the first n scores and lengths. The implementation models the reward tail and mean next length separately; it does not estimate or exploit their joint dependence.

Algorithm 1 Unknown-distribution, unknown-cost Pandora blueprint

Require: Training data $\mathcal{A}$; divisor D ; minimum openings n_0 ; horizon H

- 1: Fit a value map (if needed) and distribution/cost priors using only authorized pre-test data $\mathcal{A}$
 - 2: Open n_0 candidates and observe their scores and exact lengths
 - 3: $n \leftarrow n_0$
 - 4: **while** $n < H$ **do**
 - 5: From $\mathcal{H}_n$, construct u_n , $\widehat{F}_n^+$, and $\widehat{C}_n$
 - 6: $V_n \leftarrow \max_{j \leq n} u_n(R_j)$
 - 7: $\widehat{\text{EI}}_n \leftarrow \mathbb{E}_{X \sim \widehat{F}_n^+}[(u_n(X) - V_n)_+]$
 - 8: $G_n \leftarrow A_n \widehat{\text{EI}}_n + B_n$
 - 9: **if** $G_n \leq \widehat{C}_n/D$ **then**
 - 10: **break**
 - 11: **end if**
 - 12: Open candidate $n + 1$; observe (R_{n+1}, C_{n+1}) ; $n \leftarrow n + 1$
 - 13: **end while**
 - 14: **return** the incumbent opened candidate and charge $\sum_{j=1}^n C_j$ characters
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1.4 Alignment instantiation: exponentiated-tail UCB–Pandora with BT utility

What is fit. Each generator contains 100 prompts with 960 generations per prompt. Within each of ten split IDs 30–39, 50 prompts are training prompts and 50 are test prompts. For training prompt i , transform every raw reward as $Y_{ij} = \exp(\text{clip}(R_{ij}, -50, 50))$ and compute

$$a_i = Q_{0.5}(Y_{i1}, \dots, Y_{i,960}), \quad (4)$$

$$s_i = \max\{\text{mean}_{j:Y_{ij} \geq a_i} Y_{ij} - a_i, \epsilon\}, \quad (5)$$

$$\rho = \text{median}_{i \in \mathcal{A}} \frac{s_i}{\max(a_i, \epsilon)}. \quad (6)$$

Only the relative tail-scale prior ρ in Eq. (6) affects the retained policy. Other stored prior fields are inactive under the retained settings and are intentionally omitted. The policy configuration itself was frozen before these evaluated splits:

$$q = 0.5, \quad \kappa_r = 5, \quad \kappa_c = 0, \quad \alpha = 0.4, \quad \delta = 0.05, \quad \eta = 0.002, \quad n_0 = 3. \quad (7)$$

Here and below, $\epsilon = 10^{-8}$.

Online fit and utility. At prefix n , let T_n contain the largest $m_n = n - \lceil qn \rceil + 1$ values among $Y_1, \dots, Y_n$. The inclusive threshold convention is deliberate. Define

$$\ell_n = \min T_n, \quad \widetilde{s}_n = \max\{\text{mean}(T_n) - \ell_n, \epsilon\}, \quad (8)$$

$$s_n = \frac{m_n \widetilde{s}_n + \kappa_r \ell_n \rho}{m_n + \kappa_r}, \quad h_n = \sqrt{\frac{\log(1/\delta)}{m_n}}, \quad (9)$$

$$s_n^+ = s_n(1 + \alpha h_n), \quad s_n^- = \max\{s_n(1 - \alpha h_n), \epsilon\}. \quad (10)$$

The fitted future upper tail has mass $1 - q = 0.5$ and $Y = \ell_n + s_n^+ Z$ for $Z \sim \text{Exp}(1)$. The online proxy for the raw-reward 99th percentile and its BT utility are

$$b_n = \log(\ell_n + s_n^- \log 50), \quad u_n(r) = \sigma(r - b_n), \quad \sigma(t) = \frac{1}{1 + e^{-t}}. \quad (11)$$

Thus $u_n(r)$ is the BT probability that reward r beats the estimated high-quality benchmark, not a probability calibrated from correctness labels.

The implementation approximates the tail expected improvement with 48 fixed midpoint exponential quantiles.

Algorithm 2 Alignment: retained exponentiated-tail UCB–Pandora policy

Require: Frozen ρ from Eq. (6); divisor D ; at most 960 generations

- 1: Open the first three generations; set $n \leftarrow 3$
 - 2: **while** $n < 960$ **do**
 - 3: $Y_j \leftarrow \exp(\text{clip}(R_j, -50, 50))$ for $j \leq n$
 - 4: Form T_n from the largest $n - \lceil 0.5n \rceil + 1$ values
 - 5: Compute (ℓ_n, s_n^+, s_n^-) by Eq. (10) with $(\kappa_r, \alpha, \delta) = (5, 0.4, 0.05)$
 - 6: Compute b_n , the 48-point $\widehat{\text{EI}}_n$, G_n , and $\widehat{C}_n$ by Eqs. (11)–(13)
 - 7: **if** $G_n \leq \widehat{C}_n/D$ **then**
 - 8: **break**
 - 9: **end if**
 - 10: Open generation $n + 1$ and observe (R_{n+1}, C_{n+1}) ; $n \leftarrow n + 1$
 - 11: **end while**
 - 12: **return** the opened generation with largest raw reward and charge its prefix’s exact character sum
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For $z_k = -\log(1 - (k + 1/2)/48)$, $k = 0, \dots, 47$, and raw incumbent $r_n^* = \max_{j \leq n} R_j$,

$$\widehat{\text{EI}}_n = \frac{0.5}{48} \sum_{k=0}^{47} [\sigma(\log(\ell_n + s_n^+ z_k) - b_n) - \sigma(r_n^* - b_n)]_+, \quad (12)$$

$$G_n = \min \left\{ \widehat{\text{EI}}_n + \eta \sqrt{\frac{\log(1/\delta)}{2n}}, 1 \right\}, \quad \widehat{C}_n = \frac{1}{n} \sum_{j=1}^n C_j. \quad (13)$$

The cost prior has zero weight, so $\widehat{C}_n$ is exactly the running mean of opened character counts. Algorithm 2 is the exact retained stopping calculation.

How alignment is evaluated. For evaluation only, prompt i has benchmark $b_i^* = R_{i,(951)}$, the 951st smallest of its 960 rewards (the code’s fixed empirical 0.99-quantile convention), and stopped quality

$$Q_i(r) = \sigma(r - b_i^*). \quad (14)$$

This full-pool quantity never enters Algorithm 2; the algorithm uses the prefix benchmark b_n from Eq. (11).

1.5 Coding instantiation: shifted exponential in calibrated probability space

The coding experiments retain 83 problems for which at least one of 512 generations is correct. Each outer split contains 41 training problems and 42 test problems.

Train-only value calibration. For outer split s , one nondecreasing isotonic regression is fit directly to all 41×512 raw-reward/correctness pairs:

$$\widehat{g}_s = \text{IsoReg}\{(R_{ij}, Y_{ij}) : i \in \mathcal{A}_s, j = 1, \dots, 512\}, \quad P_{ij} = \widehat{g}_s(R_{ij}) \in [0, 1]. \quad (15)$$

Here $Y_{ij} \in \{0, 1\}$ denotes correctness. Samples have equal weight; there is no reward binning. Predictions outside the training score range are clipped to the endpoint probabilities. The frozen map is applied to both training and test rewards, and Pandora utility is simply $u(R) = \widehat{g}_s(R)$. There is no BT transform in the retained coding results.

Distribution and character priors. For each prior-fitting problem i and tail quantile q , define

$$\ell_{i,q} = Q_q(P_{i1}, \dots, P_{i,512}), \quad (16)$$

$$s_{i,q} = \max\{\text{mean}_{j:P_{ij} \geq \ell_{i,q}}(P_{ij} - \ell_{i,q}), \epsilon\}. \quad (17)$$

The prior parameters are equal-problem averages

$$\bar{\ell}_q = \text{mean}_i \ell_{i,q}, \quad \bar{s}_q = \text{mean}_i s_{i,q}, \quad \bar{C} = \text{mean}_{i,j} C_{ij}. \quad (18)$$

During four-fold policy selection, Eq. (18) is refit on the three fit folds; for final evaluation it is refit on all 41 outer-training problems. The isotonic map in Eq. (15), however, is fit once on all 41 outer-training problems before those inner folds. The final 42-problem outer test remains untouched by every fit and selection decision.

Algorithm 3 Coding: shifted-exponential Pandora in correctness-probability space

Require: Frozen isotonic map $\widehat{g}_s$; prior $(\bar{\ell}_q, \bar{s}_q, \bar{C})$; parameters $(q, \kappa_r, \kappa_c, \alpha, \gamma, \delta, D, H)$

- 1: Open three generations; observe (R_j, C_j) and set $P_j \leftarrow \widehat{g}_s(R_j)$ for $j = 1, 2, 3$; $n \leftarrow 3$
 - 2: **while** $n < H$ **do**
 - 3: Compute B_n , the local tail statistics, and (ℓ_n^+, s_n^+) by Eq. (24)
 - 4: Compute $\widehat{\text{EI}}_n$, $\widehat{C}_n$, and G_n by Eqs. (25)–(27)
 - 5: **if** $G_n \leq \widehat{C}_n/D$ **then**
 - 6: **break**
 - 7: **end if**
 - 8: Open generation $n + 1$; observe (R_{n+1}, C_{n+1})
 - 9: $P_{n+1} \leftarrow \widehat{g}_s(R_{n+1})$; $n \leftarrow n + 1$
 - 10: **end while**
 - 11: **return** the first opened generation attaining $\max_{j \leq n} P_j$ and charge the exact prefix character sum
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Online tail fit. For an opened prefix $P_1, \dots, P_n$, let

$$B_n = \max_{j \leq n} P_j, \quad \ell_n = Q_q(P_1, \dots, P_n), \quad (19)$$

$$I_n = \{j \leq n : P_j \geq \ell_n\}, \quad m_n = |I_n|, \quad (20)$$

$$s_n = \max \left\{ \frac{1}{m_n} \sum_{j \in I_n} (P_j - \ell_n), \epsilon \right\}. \quad (21)$$

Location and scale are shrunk asymmetrically toward the training prior:

$$\tilde{\ell}_n = \frac{n\ell_n + \kappa_r \bar{\ell}_q}{n + \kappa_r}, \quad \tilde{s}_n = \frac{m_n s_n + \kappa_r \bar{s}_q}{m_n + \kappa_r}, \quad (22)$$

$$h_n = \sqrt{\frac{\log(1/\delta)}{m_n}}, \quad \ell_n^+ = \text{clip}_{[0,1]}(\tilde{\ell}_n + \alpha \tilde{s}_n h_n), \quad (23)$$

$$s_n^+ = \tilde{s}_n(1 + \alpha h_n). \quad (24)$$

The future upper tail has fixed mass $1 - q$. Conditional on being in that tail, X_n follows a shifted exponential with location ℓ_n^+ and scale s_n^+ , truncated and renormalized on $[\ell_n^+, 1]$:

$$f_n(x) = \frac{\exp(-(x - \ell_n^+)/s_n^+)}{s_n^+ \{1 - \exp[-(1 - \ell_n^+)/s_n^+]\}}, \quad \ell_n^+ \leq x \leq 1. \quad (25)$$

Consequently,

$$\widehat{\text{EI}}_n = (1 - q) \int_{\ell_n^+}^1 (x - B_n)_+ f_n(x) dx, \quad (26)$$

$$\widehat{C}_n = \frac{\sum_{j=1}^n C_j + \kappa_c \bar{C}}{n + \kappa_c}, \quad G_n = \left(\frac{n}{3}\right)^{-\gamma} \widehat{\text{EI}}_n. \quad (27)$$

The code evaluates the integral using 32-point Gauss–Legendre quadrature through the inverse CDF of Eq. (25). Specifically, if $(x_k, w_k)_{k=1}^{32}$ are the Gauss–Legendre nodes and normalized weights on $[0, 1]$, and $M_n = 1 - \exp[-(1 - \ell_n^+)/s_n^+]$, the computed value is

$$\widehat{\text{EI}}_n = (1 - q) \sum_{k=1}^{32} w_k [\ell_n^+ - s_n^+ \log(1 - M_n x_k) - B_n]_+. \quad (28)$$

At $\ell_n^+ = 1$ the conditional law is interpreted as a point mass at one. Its tail mass remains exactly $1 - q$ even when isotonic plateaus make m_n larger than $(1 - q)n$.

Equation (24) is optimistic only when $\alpha > 0$. If $\alpha = 0$, Algorithm 3 is a prior-regularized plug-in Pandora rule, not a UCB rule. The multiplier $(n/3)^{-\gamma}$ and the optional horizon $H < 512$ are empirical regularizers rather than parts of classical Pandora stopping. Isotonic plateaus can tie several candidates; the retained implementation returns the first one attaining the maximum calibrated probability.

Table 1: Exact instantiations of the shared blueprint. “Prefix” means that the quantity is recomputed using only opened candidates.

Component	Alignment	Coding
Tail-model space	$\exp(R)$, unbounded shifted-exponential upper tail	$\hat{g}_s(R) \in [0, 1]$, shifted-exponential upper tail truncated at 1
Decision utility	Prefix BT probability $\sigma(R - b_n)$	Frozen isotonic correctness probability $\hat{g}_s(R)$
Active training fit	Median tail scale/location ratio ρ	Isotonic map, tail location/scale priors, and mean characters
Optimism	Upper tail scale, lower benchmark scale, plus an EI bonus	Optional upper location and scale controlled by α
Next-cost estimate	Mean characters in the opened prefix ($\kappa_c = 0$)	Prefix/prior shrinkage with $\kappa_c \in \{0, 10\}$
EI regularizer	Additive confidence bonus	Multiplicative $(n/3)^{-\gamma}$ tail decay
Hard horizon	960, the available pool	512 or a training-chosen cap in the utility experiment; 512 for targets

1.6 What each experiment measures

1.6.1 Alignment experiments

The five alignment generators are Gemma-2-9B, Llama-3.1-8B, Llama-3.2-3B, Mistral-7B, and Qwen-2.5-7B. Each split uses four randomized training permutations and eight randomized test permutations per prompt. The three experiments reuse the same stopped prefixes but answer different questions:

1. **Utility objective.** At each reported divisor $D \in \{10^5, 5 \times 10^5, 10^6, 5 \times 10^6, 10^7, 5 \times 10^7\}$, training data choose the integer N maximizing $\text{mean}(Q) - \text{mean}(C)/D$. That N is frozen on test, and its utility is compared with Algorithm 2.
2. **Equal-character head-to-head.** On an independent set of test permutations, a randomized mixture of two adjacent Fixed- N policies is chosen so that its *aggregate expected* character count exactly matches the adaptive mean. The reported score is $\Pr(R_{\text{adaptive}} > R_{\text{fixed}}) + \frac{1}{2} \Pr(\text{tie})$. Because the matching uses held-out outcomes, this is a descriptive equal-budget comparison, not a deployable selection rule.
3. **Target quality.** For requested BT qualities $0.30, 0.35, \dots, 0.60$, training interpolation selects between adjacent divisors on an 11-point adaptive frontier; a parallel interpolation selects adjacent Fixed- N values. Both are frozen for test. The main cost-saving curve then performs a second, post-hoc interpolation of the held-out Fixed- N curve to the adaptive policy’s achieved held-out quality. This quality-matched baseline is an oracle used for description, whereas the separately recorded direct comparison to the train-selected Fixed- N rule is deployable. Twenty-eight of 350 split-target rows, almost all at target 0.30, lie outside the training adaptive frontier and are clamped.

Figure 1 shows the three questions side by side. Across $D \leq 10^7$, relative utility gains range from 1.31% to 15.19% and all 25 generator-divisor confidence intervals are above zero. At $D = 5 \times 10^7$ the gains are only 0.08–0.33%, and three of five intervals include zero. Every equal-character head-to-head interval is above 0.5. Quality-matched target savings range from 10.23% to 37.52%, with all reported intervals above zero; these latter savings retain the oracle-matching caveat.

1.6.2 Coding utility and equal-budget experiments

For split IDs 75–84, each policy is tuned on 41 training problems using four folds and four permutations per problem, then evaluated on 42 untouched test problems with 48 permutations each. For every divisor, selection is restricted to the shifted-exponential family

$$\begin{aligned}
 q \in \{0.5, 0.75\}, \quad \alpha \in \{0, 0.2, 0.8\}, \quad \kappa_r \in \{5, 20, 10^6\}, \\
 \kappa_c \in \{0, 10\}, \quad a \in \{1, 1.1, 1.2, 1.3, 1.5, 2, \infty\}, \quad \gamma = 0.
 \end{aligned} \tag{29}$$

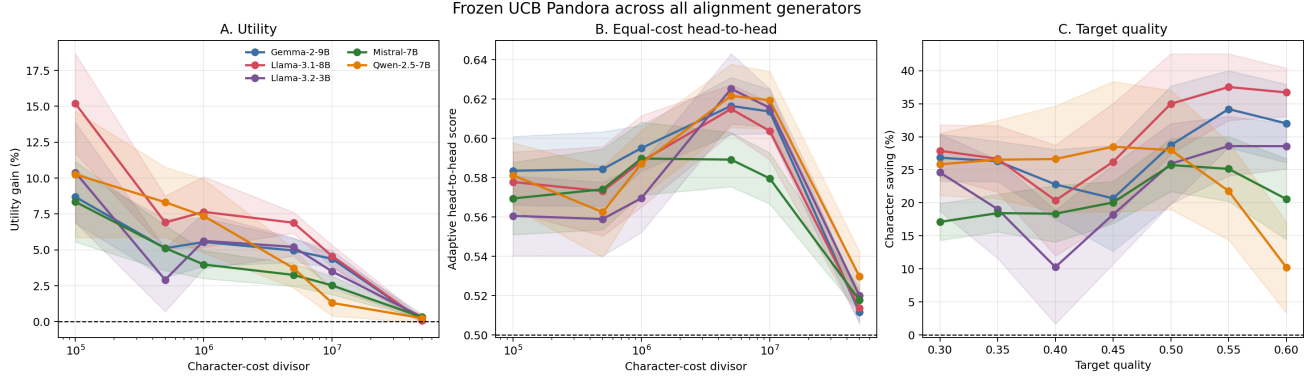


Figure 1: Alignment results over ten prompt splits for five generators. **A**: relative utility gain over the train-tuned, frozen Fixed- N policy. **B**: raw-reward head-to-head score against an independent adjacent- N mixture with exactly matched expected held-out characters; 0.5 is parity. **C**: character reduction relative to a post-hoc held-out Fixed- N interpolation matched to achieved quality. Bands are 95% Student- t intervals over ten splits. Panels B and C are controlled descriptive comparisons, not policy-selection procedures.

The selected configuration maximizes four-fold training utility accuracy – characters $/D$. If $a < \infty$, its horizon is

$$H = \min\{512, \max(3, \lceil aN_D^* \rceil)\}, \quad N_D^* = \arg \max_N \left\{ A_{\text{train}}(N) - \frac{N\bar{C}}{D} \right\}; \quad (30)$$

otherwise $H = 512$. Thus this experiment is not uniformly uncapped: 91 of the 100 saved split-divisor policies use a finite horizon. Among the same 100 selections, 56 use $\alpha > 0$ and 44 use $\alpha = 0$.

The *utility-gap* view compares test utility with the train-tuned, frozen N_D^* . The *equal-budget accuracy* view uses independent test permutations and a post-hoc adjacent- N mixture whose mean characters exactly match the adaptive mean. It isolates accuracy at a common measured budget but is not deployable because the mixture is obtained from held-out costs. For example, at $D = 10^5$ the shifted-exponential policy gains 5.75% utility (95% CI 2.81–8.69%) and 1.51 accuracy points at equal characters (0.55–2.47). At $D = 3 \times 10^5$, the corresponding gains are 1.85% (0.09–3.61) and 0.73 points (0.17–1.28). Most higher-divisor utility intervals include zero, so the results do not support a claim of uniform gain at every price.

1.6.3 Coding target-quality experiment

For final split IDs 95–104, four-fold selection uses 16 training permutations per problem and final evaluation uses 48 test permutations per problem. Each target $\tau \in \{0.25, 0.30, 0.35\}$ selects one uncapped shifted-exponential configuration and one of 19 divisors. The family grid is

$$q \in \{0.5, 0.75\}, \quad \alpha \in \{0, 0.2, 0.8\}, \quad \kappa_r \in \{5, 20, 10^6\}, \quad \kappa_c \in \{0, 10\}, \quad \gamma \in \{0, 0.25, 0.5, 1\}, \quad H = 512. \quad (31)$$

A frozen development profile is first built from held-out evaluations on split IDs 55–59. For policy h and divisor D , it stores mean accuracy $A_{\text{prof}}(h, D)$ and geometric-mean characters

$$C_{\text{prof}}(h, D) = \exp \left\{ \frac{1}{5} \sum_{s=55}^{59} \log C_s(h, D) \right\}. \quad (32)$$

On current outer split s , the feasibility score is

$$A_{\text{sel},s}(h, D) = 0.1A_{\text{CF},s}(h, D) + 0.9A_{\text{prof}}(h, D). \quad (33)$$

Among candidates with $A_{\text{sel},s} \geq \tau$, selection minimizes frozen C_{prof} , with smaller overshoot and then fixed identifiers breaking ties. Current-split character cost receives zero selection weight. The implementation has a cheapest-max-score fallback if no candidate is feasible, but it was inactive because all 30 retained choices passed their selection threshold. There is no policy mixture, target margin, lower-confidence threshold, Fixed- N guard, or test-time selection. All 30 selected policies have $\gamma > 0$; 27 have $\alpha > 0$ and three have $\alpha = 0$.

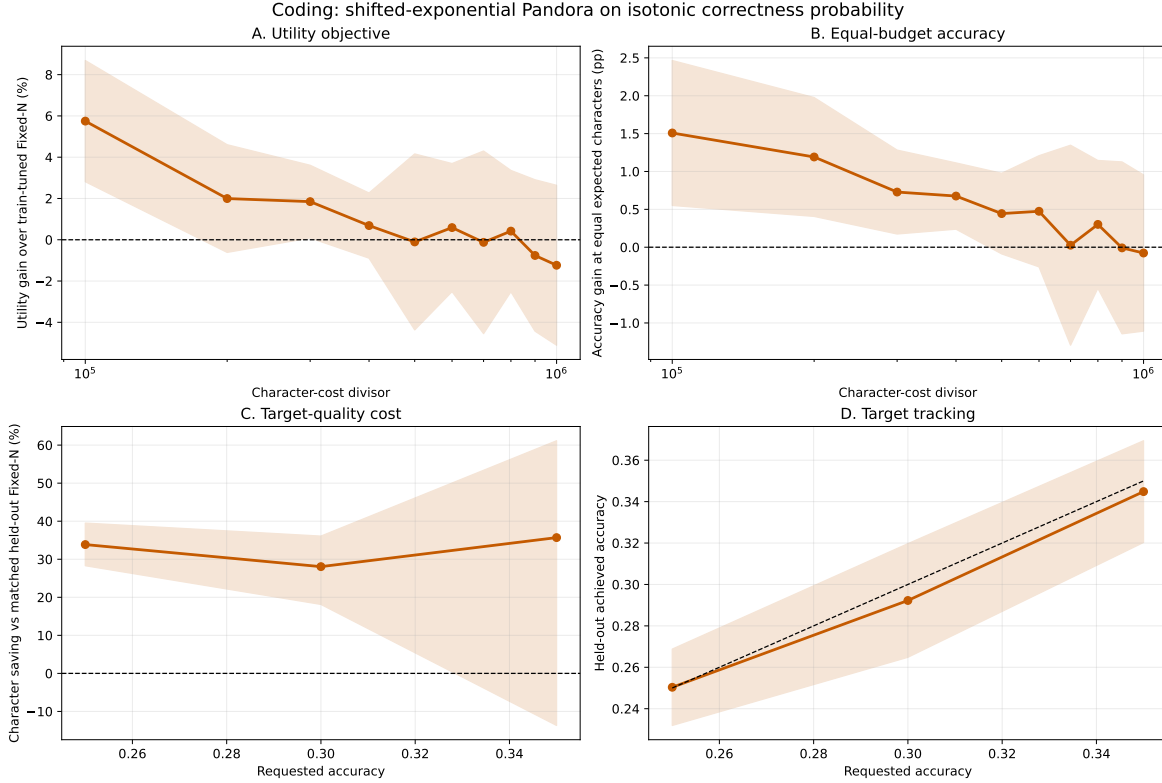


Figure 2: Coding results for the shifted-exponential calibrated-probability family only. **A:** relative test-utility gain over train-tuned Fixed- N . **B:** test-accuracy gain against a post-hoc adjacent- N mixture with exactly matched expected characters. **C:** target-mode character savings against a post-hoc held-out Fixed- N matched to achieved accuracy. **D:** achieved versus requested accuracy; the diagonal is exact target tracking. Utility/equal-budget and accuracy bands are 95% Student- t intervals over ten splits; panel C uses a paired split bootstrap. Panels B and C are descriptive oracle matches, while panels A and D evaluate policies selected without final-test outcomes.

Two integer Fixed- N baselines answer different questions. The deployable baseline is the cheapest training N whose exact training accuracy reaches τ , frozen before test. The descriptive oracle is the cheapest held-out N whose accuracy reaches the *adaptive policy's achieved held-out accuracy*. Mean achieved accuracies for requested targets 0.25, 0.30, and 0.35 are 0.2503, 0.2923, and 0.3448, respectively. The respective character savings against the held-out matched oracle are 33.86%, 28.06%, and 35.69%; the last 95% interval includes zero. Per-split target hit rates are 5/10, 3/10, and 3/10, so the procedure tracks targets in aggregate but does not guarantee that every held-out split reaches them.

1.7 Relation to unknown-distribution, unknown-cost Pandora's box

Both systems are faithful instantiations of the same empirical Pandora blueprint: opening a generation reveals an initially unknown value and an initially unknown character cost; a training-regularized tail law estimates the value of one more opening; an online mean estimates its cost; and the policy stops when adjusted expected improvement no longer pays for that cost. Alignment instantiates the blueprint with a genuinely optimistic exponentiated tail and a dynamic BT utility. Coding instantiates it with a bounded shifted-exponential tail after a frozen isotonic utility transform, plus optional optimism, optional character shrinkage, tail decay, and—in the utility experiment—often a training-derived hard horizon.

Accordingly, *unknown-distribution, unknown-cost Pandora policy* is the accurate umbrella description. *UCB-Pandora* is accurate for the alignment rule and for coding configurations with $\alpha > 0$, but not for coding configurations with $\alpha = 0$. These are empirical reservation policies, not claims of Weitzman-optimal behavior: the tail model can be misspecified, cost uses only a marginal mean, coding adds empirical regularizers, and finite pools sampled without replacement impose hard horizons. These qualifications identify each instantiation without changing the blueprint in Algorithm 1.